# B4.1 Adaptation to environment

*Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/sl-b4-1.html*

*Written for this site from the class slides, with textbook page references for each statement. The chapter is on pages 341–362 of the course textbook. Image credits are on the last slide of the slides for this subtopic.*

## HABITATS AND ADAPTATIONS

### B4.1.1 Habitat

pp. 341–342

A **habitat** is the place where an organism, population, species or community lives. A description can include geographical and physical location, and the type of ecosystem.

### B4.1.2 Adaptations to the abiotic environment

pp. 342–344

| PLANT | CHALLENGES | ADAPTATIONS |
|---|---|---|
| MARRAM GRASS, SAND DUNES | Little water, salt spray, wind | Rolled leaves, sunken stomata, thick cuticle, deep roots and rhizomes |
| MANGROVE TREES, SWAMPS | Anaerobic, salty, soft mud | Pneumatophores, stilt roots, salt glands |

### B4.1.3 Abiotic variables and distribution

pp. 344–345

Plants are affected by temperature, water, light, soil pH, salinity and mineral nutrients. Animals are affected by temperature, water and breeding sites, among others. A species' adaptations give it a range of tolerance.

### B4.1.4 Range of tolerance

pp. 345–352

Each species thrives in an **optimum** range, is under stress towards the limits, and cannot survive beyond them. A **transect** lets you correlate the distribution of a species with an abiotic variable, using data you collect yourself. A correlation alone does not prove cause.

## B4.1.5 Conditions for coral reefs

p. 353

### Conditions for coral reefs

B4.1.5

| Factor | Condition | Why |
|---|---|---|
| Depth | Mostly under 25 m | Light reaches algae |
| pH | Above 7.8 | Calcium carbonate can form |
| Salinity | About 35 parts per thousand | Avoids osmotic stress |
| Clarity | Clear water | Light passes through |
| Temperature | About 23–29 °C | Suits corals and algae |

**Threats**
Warming causes bleaching; ocean acidification lowers pH.

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Reefs need shallow, clear, warm water with suitable pH and salinity. Most reef-building corals live less than 25 m deep, at about 23–29 °C. Corals have a mutualistic relationship with algae, the zooxanthellae, which need light.

## B4.1.6 Climate determines land biomes

pp. 354–355

For any given temperature and rainfall pattern, one type of natural ecosystem is likely to develop. A graph of these two variables shows where each biome is found.

## B4.1.7 Similar communities in each biome

pp. 356–359

Similar abiotic conditions produce similar selection pressures, and so **convergent evolution** gives similar communities. Know the climates of tropical forest, temperate forest, taiga, grassland, tundra and hot desert.

## B4.1.8 Hot deserts and tropical rainforests

pp. 359–361

| SPECIES | BIOME | ADAPTATIONS |
|---|---|---|
| SAGUARO CACTUS | Hot desert | Water-storing stem, spines, shallow wide roots |
| FENNEC FOX | Hot desert | Large ears lose heat, active at night |
| SPIDER MONKEY | Rainforest | Long limbs and a gripping tail |
| POISON DART FROG | Rainforest | Toxins and warning colours |

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